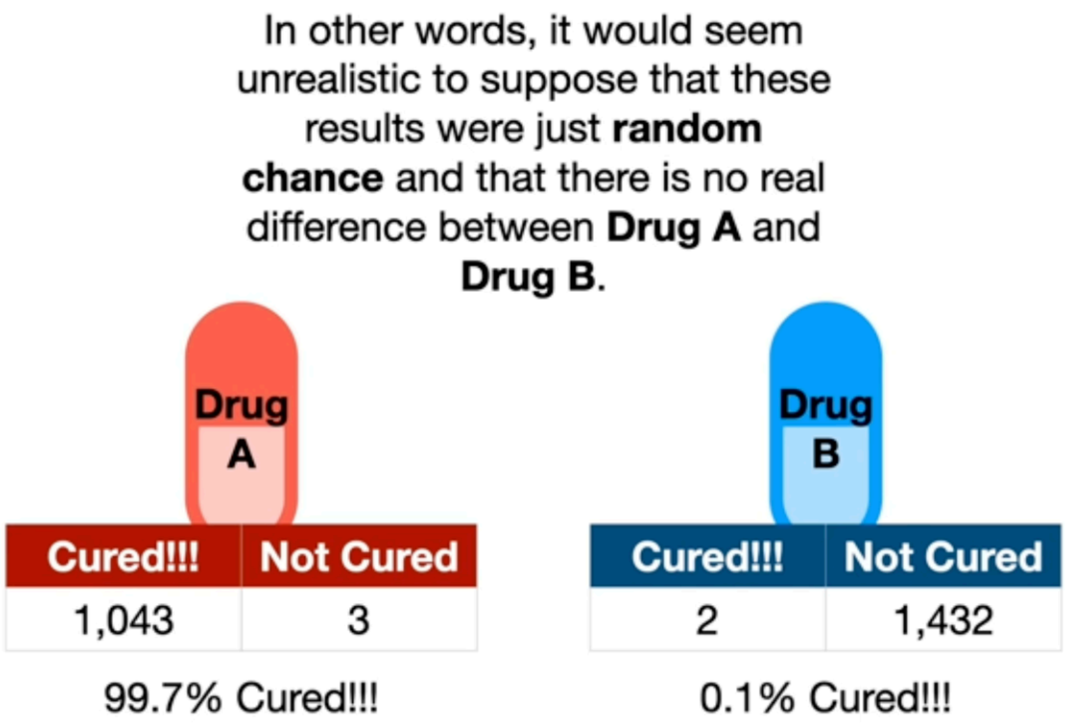
p-value v1

the 1 person using drug A is cured.
the 2 person using drug B is not cured. (there is a lot of thing happened that affect the result)
so test it on a lot people would be considerable

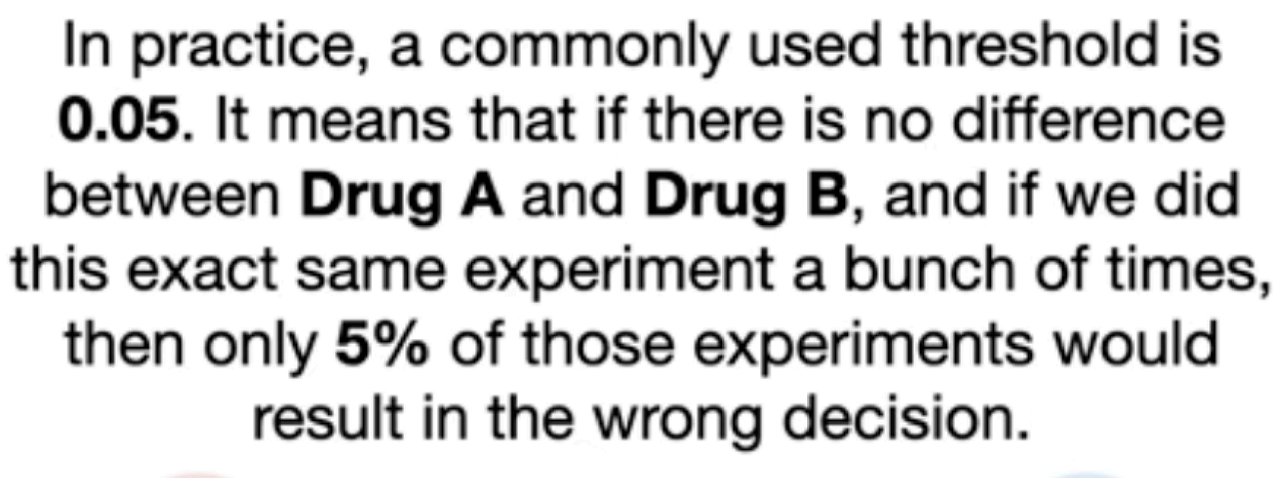


So P-value can solve this problem

P-values are numbers between 1 and 0, that, in this example, quantify how confident we should be that Drug A is different from Drug B

help us to decide whether or not to reject the null hypothesis.

The closer a P-value is to 0, the more confidence we have that Drug A and Drug B are different.



Getting a small P-Value when there is no difference is called a False positive.

A **0.05** threshold for **p-values** means that **5%** of the experiments, where the only differences come from weird random things, will generate a **p-value** smaller than **0.05**.

In other words, if there is no difference between **Drug A** and **Drug B**, **5%** time we do the experiment, we'll get a **p-value** less than **0.05**, aka a **False Positive**.

NOTE: If it is extremely important that we are correct when we say the drugs are different, then we can use a smaller threshold, like **0.00001**.

Using a threshold of **0.00001** means we would only get a **False Positive** once every **100,000** experiments.

In summary, a small **p-value** does not imply that the **effect size**, or difference between **Drug A** and **Drug B**, is large.